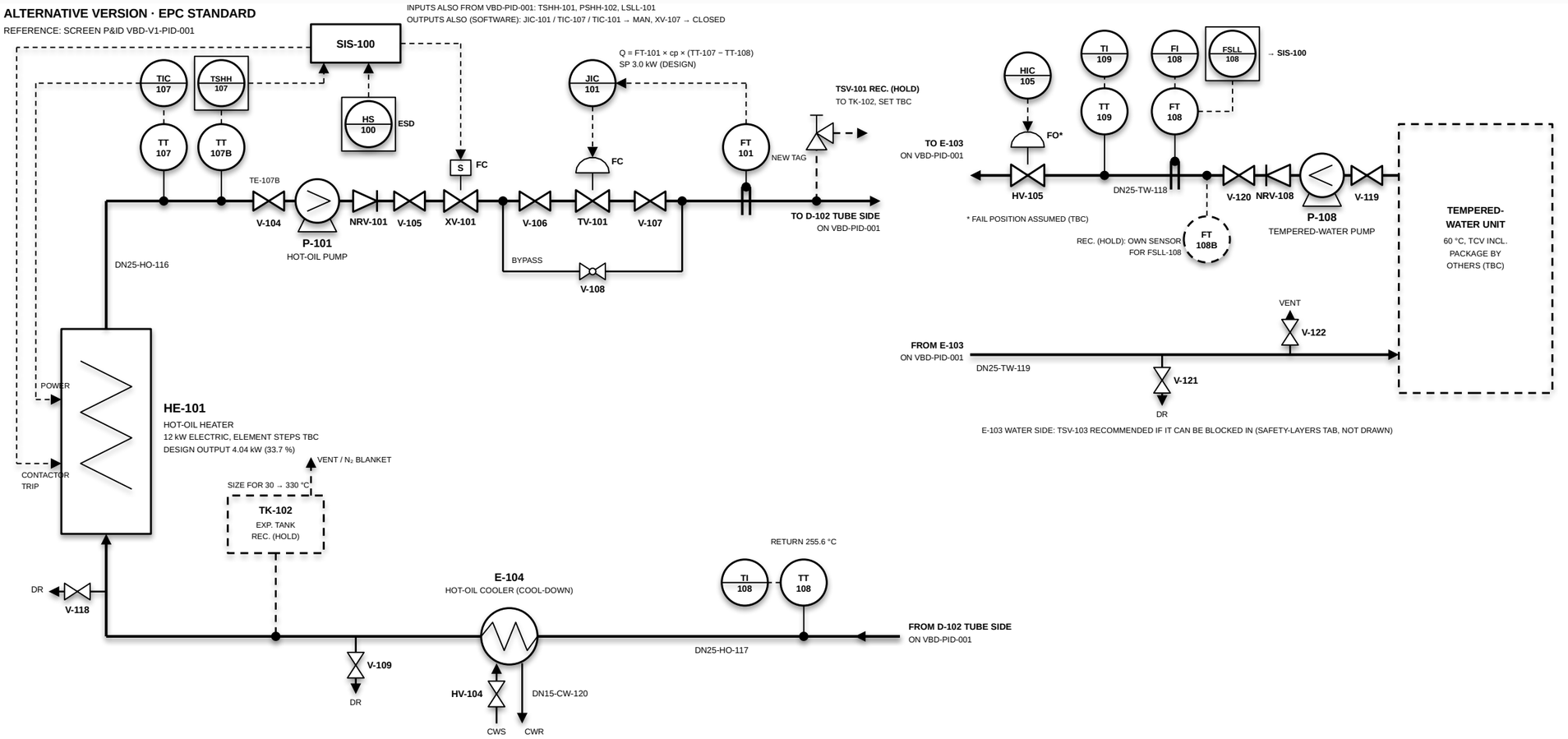
ALTERNATIVE VERSION · EPC STANDARD

REFERENCE: SCREEN P&ID VBD-V1-PID-001



E-103 WATER SIDE: TSV-103 RECOMMENDED IF IT CAN BE BLOCKED IN (SAFETY-LAYERS TAB, NOT DRAWN)

INTERLOCK SCHEDULE · SIS-100 AND P-104 PUMP INTERLOCK (FROM THE OTS MODEL)

X = ACTION ON TRIP. SETTINGS FROM K.trip*; DELAYS FROM sisStep [5, 5, 10, 5, 5] s; ACTIONS FROM sisTrip.

ITEM	TAG	SENSOR	CAUSE	SETTING	DELAY	ACTIVE	TV-101 SHUT (JIC-101 MAN 0 %)	HE-101 OFF (TIC-107 MAN 0 %)	XV-101 CLOSED	TIC-101 → MAN	XV-107 → CLOSED (TOT. REFL.)	P-104 STOP	FIRST-OUT + ALARM	RESET
I-1	TSHH-101	TE-101B / TT-101B (OWN)	POT D-101 TEMPERATURE HIGH-HIGH	> 290 °C	5 s	ALWAYS	X	X	X	X	X		X	BLOCKED UNTIL POT < 290 °C
I-2	PSHH-102	PT-102 (SHARED, PIC-102)	COLUMN TOP PRESSURE HIGH-HIGH	> 300 mbar	5 s	HEATING ARMED	X	X	X	X	X		X	ANY TIME; RE-ARM NEEDS PERMISSIVES
I-3	FSLL-108	FT-108 (SHARED, FI-108)	TEMPERED WATER TO E-103 LOW-LOW	< 0.36 m³/h	10 s	HEATING ARMED	X	X	X	X	X		X	ANY TIME; RE-ARM NEEDS PERMISSIVES
I-4	LSLL-101	LT-101 (SHARED, LI-101)	POT D-101 LEVEL LOW-LOW	< 20 %	5 s	HEATING ARMED	X	X	X	X	X		X	ANY TIME; RE-ARM NEEDS PERMISSIVES
I-5	TSHH-107	TE-107B / TT-107B (OWN)	HOT-OIL SUPPLY TEMPERATURE HIGH-HIGH	> 330 °C	5 s	ALWAYS	X	X	X	X	X		X	BLOCKED UNTIL OIL < 330 °C
I-6	HS-100	PUSH-BUTTON (DCS / SIS)	MANUAL EMERGENCY STOP	PUSH	0 s	ALWAYS	X	X	X	X	X		X	RESET, THEN RE-ARM
I-7	LSLL-104	LT-104 (SHARED, LIC-104)	REFLUX DRUM LEVEL LOW-LOW (BPCS, NOT A SIF)	< 5 %	5 s	P-104 RUNNING						X	ALARM	RESTART P-104 IN THE FIELD
REC.	TZ-110	TT-101B (REC., HOLD)	N_2 VACUUM BREAK HV-110 WITH POT > 150 °C	> 150 °C	—	ALWAYS								BLOCKS HV-110 (NOT IN MODEL)

PERMISSIVES TO ARM HEATING (armPermissives, ALL REQUIRED): VP-101 RUNNING AND PIC-102 < 250 mbar · P-108 RUNNING AND FI-108 ≥ 0.6 m³/h · LI-101 > 30 % · P-101 RUNNING · TIC-107 < 320 °C · HV-110 CLOSED. ARMING OPENS XV-101 AND ENABLES HE-101. OPERATOR DISARM: XV-101 CLOSED, HE-101 OFF.

VP-101, P-108, P-101 AND P-104 KEEP RUNNING ON A SIS TRIP: VACUUM, CONDENSER AND REFLUX STAY ON SO THE COLUMN COOLS UNDER TOTAL REFLUX.

LEGEND

- HAND VALVE (BLOCK / DRAIN), V-1nn
- CHECK VALVE, FLOW →, NRV-1nn
- GLOBE (BYPASS) VALVE
- CONTROL VALVE, DIAPHRAGM; FO / FC = FAIL OPEN / CLOSED
- ON/OFF VALVE, SOLENOID-PNEUMATIC (S)
- THREE-WAY DIVERTER VALVE
- RELIEF VALVE (DASHED = RECOMMENDED, HOLD)
- ORIFICE FLOW ELEMENT
- CENTRIFUGAL PUMP (MOTOR NOT SHOWN)
- FIELD INSTRUMENT (TRANSMITTER / ELEMENT)
- DCS FUNCTION (SHARED DISPLAY / CONTROL)
- SIS-100 FUNCTION (TRIP)
- RECOMMENDED INSTRUMENT (DASHED, HOLD)
- INTERLOCK LOGIC (DIAMOND)
- INSTRUMENT SIGNAL (ELECTRIC / SOFTWARE)
- PROCESS LINE, FLOW DIRECTION
- UTILITY / DRAIN / VENT LINE
- CONNECTION (TEE)

LINE NUMBER DNxx-SVC-1nn (SIZES PRELIMINARY, TBC)

PV PROCESS VAPOUR · PL PROCESS LIQUID · VG VACUUM GAS
NG NITROGEN · HO HOT OIL · TW TEMPERED WATER · CW COOLING WATER

RECOMMENDED ADDITIONS (DASHED, NOT EXISTING, HOLD)

PSV-101 C-101 TOP · RD-102 + PSV-102 POT (PID-001)
TSV-101 HOT-OIL SIDE OF D-102 → TK-102 (PID-002)
TK-102 HOT-OIL EXPANSION TANK, N_2 BLANKET (PID-002)
PT-102B, LT-101B (PID-001), FT-108B (PID-002):
OWN SENSORS FOR PSHH-102, LSLL-101, FSLL-108
TZ-110 HV-110 INTERLOCK, POT > 150 °C (PID-001)
SOURCE: HAZOP R02/R03/R13 AND SAFETY-LAYERS REGISTER

NOTES (HOT OIL AND TEMPERED WATER)

1. PART OF THE ALTERNATIVE VERSION (EPC STANDARD). GENERAL NOTES, EQUIPMENT LIST AND PROCESS SIDE: VBD-PID-001.
2. HOT OIL: SUPPLY 290 °C (TIC-107), RETURN 255.6 °C (ΔT 34.4 K). TV-101 36.7 % OPEN GIVES mcp 0.087 kW/K = 125 kg/h AT cp 2.5 kJ/kg K (ASSUMED); TV-101 WIDE OPEN ≈ 1080 kg/h. HE-101 4.04 kW = POT DUTY 3.00 kW + LOOP LOSS 1.04 kW (BALANCE CLOSES TO 0.000 kW).
3. OIL GRADE, FLASH POINT, LOOP VOLUME, P-101 HEAD, HE-101 ELEMENT STEPS: TBC. HE-101 HAS POWER ONLY WITH P-101 RUNNING (MODEL; CONFIRM).
4. JIC-101 DUTY = FT-101 × cp × (TT-107 − TT-108). FT-101 IS A NEW TAG FOR THE FLOW THE MODEL USES; METER TYPE AND RANGE TBC.
5. TEMPERED WATER 60 °C, 1.2 m³/h; ΔT ACROSS E-103 1.7 K AT 2.37 kW. KEEP SUPPLY ABOVE 45 °C (TI-109 L ALARM): ESTERS FREEZE AT 30–38 °C.
6. E-104 / HV-104 ONLY FOR COOL-DOWN (SHUTDOWN PROCEDURE). COOLING-WATER CONDITIONS TBC.
7. DCS ALARMS (OTS): TIC-107 H 315 °C · JIC-101 H 5.0 kW · FI-108 L 0.6 m³/h · TI-109 L 45 / H 80 °C · TIC-101 H 280 °C.
8. LINE SIZES PRELIMINARY (TBC): HOT OIL DN25 = 0.72 m/s AT 1080 kg/h (ρ 750 ASSUMED); TEMPERED WATER DN25 = 0.60 m/s.

INTERLOCK NOTES

9. A TRIP SETS JIC-101 AND TIC-107 TO MAN 0 % (TV-101 SHUT, HE-101 OFF), TIC-101 TO MAN, XV-107 TO CLOSED; SIS-100 LEAVES RUN SO XV-101 CLOSES.
10. PSHH-102, FSLL-108 AND LSLL-101 ACT ONLY WHILE HEATING IS ARMED. TSHH-101 AND TSHH-107 ACT ALWAYS AND BLOCK THE RESET WHILE ABOVE THE SETTING.
11. SIS-100 DE-ENERGISES TO TRIP; XV-101 AND TV-101 FAIL CLOSED; HE-101 CONTACTOR OPENS. FIRST-OUT SHOWN ON THE SIS FACEPLATE.
12. SHARED SENSORS (SAFETY-LAYERS TAB GAPS): PT-102, FT-108, LT-101 SERVE CONTROL / ALARM AND TRIP. OWN SENSORS ARE RECOMMENDED (HOLD).
13. AIR INGRESS (HIGH RISK IN THE HAZOP) HAS NO SIF: TZ-110 AND AN O_2 ANALYSER AT-102 ARE RECOMMENDED (HOLD).
14. SIL TARGETS, PROOF TESTS AND RESPONSE TIMES: SRS OF SIF TAB. CAUSE AND EFFECT: C&E MATRIX VBD-IN-CE-001.

REV	DESCRIPTION	DATE
—	VBD-V1-PID-001 A (SCREEN): CURRENT, DCS/DECKS	—
A	ALTERNATIVE VERSION, EPC STANDARD	12-OCT-26

CLIENT: DEPARTMENT OF CHEMICAL ENGINEERING,
FACULTY OF ENGINEERING, CHULALONGKORN UNIVERSITY
VACUUM BATCH DISTILLATION VER 1 · OTS

P&ID — VACUUM BATCH DISTILLATION C-101 · ALTERNATIVE VERSION

DRAWN: Apinan Soottitantawat	CHK'D: --	APP'D: --	DATE: 12-OCT-26
DWG NO. VBD-PID-002	REV A	SCALE N.T.S.	SHEET 2 OF 2